

Python Programming – Comprehensive Syllabus

Course Overview

This course provides a comprehensive introduction to Python programming, covering programming fundamentals, object-oriented programming, file handling, modules, exception handling, data structures, and application development. Students will develop problem-solving skills and build real-world applications using one of the most widely used programming languages.

Duration: 8–12 Weeks (50–70 Hours)

Prerequisites:

- Basic computer knowledge
 - No prior programming experience required
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Module 1: Introduction to Python (Week 1)

Topics

- Introduction to Python
- Features of Python
- Installing Python and IDEs
- Python Syntax
- Variables and Data Types
- Operators
- Input and Output
- Type Conversion
- Comments and Naming Conventions

Practical Exercises

- Hello World Program
- Basic Calculator
- Unit Conversion Programs

Module 2: Decision Making & Looping (Week 2)

Topics

- if, elif, else Statements
- Nested Conditions
- for Loop
- while Loop
- break, continue, and pass
- Range Function

Practical Exercises

- Grade Calculator
 - Leap Year Checker
 - Multiplication Table
 - Number Guessing Game
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Module 3: Functions, Strings & Collections (Week 3)

Topics

- Functions
- Arguments and Return Values
- Lambda Functions
- Recursion
- Strings
- Lists
- Tuples
- Sets
- Dictionaries
- List Comprehensions

Practical Exercises

- String Manipulation

- Student Record Management
 - Dictionary-Based Phonebook
 - List Processing Programs
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Module 4: Modules, Packages & File Handling (Week 4)

Topics

- Importing Modules
- Creating Modules
- Python Packages
- File Handling
- Reading and Writing Files
- CSV and JSON Files
- Working with Directories

Practical Exercises

- File Management System
 - CSV Data Processing
 - JSON Data Reader
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Module 5: Object-Oriented Programming (Week 5)

Topics

- Classes and Objects
- Constructors
- Instance and Class Variables
- Inheritance
- Polymorphism
- Encapsulation
- Abstraction
- Special Methods (**init**, **str**, etc.)

Practical Exercises

- Library Management System
 - Employee Management System
 - Bank Account Simulation
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Module 6: Exception Handling & Python Libraries (Week 6)

Topics

- Exception Handling (try, except, else, finally)
- Custom Exceptions
- Regular Expressions
- Date and Time Module
- Math Module
- Random Module
- OS Module

Practical Exercises

- Error Handling Programs
 - File Validation
 - Password Validator
 - Date and Time Applications
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Module 7: Database Connectivity & GUI Development (Week 7)

Topics

- SQLite Database
- CRUD Operations
- Introduction to MySQL Connectivity
- Tkinter Basics
- GUI Widgets
- Event Handling

Practical Exercises

- Student Database Application
 - Inventory Management
 - Calculator GUI
 - Registration Form
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Module 8: Introduction to Web Development & APIs (Week 8)

Topics

- Introduction to Flask
- Routing
- Templates
- Forms
- REST APIs
- JSON Handling
- Virtual Environments
- Package Management (pip)

Practical Exercises

- Personal Portfolio Website
 - Simple CRUD Web Application
 - Weather API Integration
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Module 9: Mini Project (Week 9)

Choose one:

- Library Management System
 - Banking Management System
 - Student Information System
 - Inventory Management System
 - Employee Payroll System
 - Hospital Management System
 - Expense Tracker
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Module 10–12: Capstone Project

Complete Python Application Development

- Requirement Analysis
 - Application Design
 - Database Integration
 - GUI or Web Interface Development
 - CRUD Operations
 - Exception Handling
 - Testing and Debugging
 - Documentation
 - Final Presentation
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Tools & Software

- Python 3.x
 - Visual Studio Code
 - PyCharm
 - Jupyter Notebook
 - SQLite
 - MySQL
 - Flask
 - Git & GitHub
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Core Concepts Covered

- Python Programming Fundamentals
 - Functions and Modules
 - Strings and Collections
 - Object-Oriented Programming (OOP)
 - Exception Handling
 - File Handling
 - Database Connectivity
 - GUI Development with Tkinter
 - Introduction to Web Development with Flask
 - REST APIs and JSON
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Assessment Scheme

Component	Weightage
Assignments	20%
Lab Exercises	20%
Quizzes	10%
Mini Project	20%
Capstone Project	30%

Learning Outcomes

By the end of the course, learners will be able to:

- Understand Python syntax and programming fundamentals.
- Develop modular programs using functions, modules, and packages.
- Work effectively with strings, lists, tuples, sets, and dictionaries.
- Apply object-oriented programming concepts to build reusable applications.
- Handle exceptions and perform file operations for robust software.
- Connect Python applications to databases and implement CRUD operations.
- Develop desktop applications using Tkinter and basic web applications using Flask.
- Consume REST APIs and work with JSON data.
- Debug, test, and optimize Python applications using industry-standard tools.
- Build real-world Python applications following modern software development practices.